

# MQTT backend docking protocol interface document of facial recognition device

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- based on MQTT 3.1 communication protocol
- Document version 1.0
- Release date August 8, 2024
- Modified by Jinjie Lai

## Update Record

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| Version number | modification date | Explain |
|----------------|-------------------|---------|
|                |                   |         |

## MQTT protocol definition

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- Support MQTT 3.1
- GZIP compression technique
- Support TLS1.2 TLS1.3

## MQTT Topic Definition

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- Device subscription subject
  - /iot\_hub/publish/{SN}
  - Example /iot\_hub/publish/FC-8200H12345678
  - Messages sent from the server to the device should be sent under this topic
- Topic of device message release
  - /iot\_hub/upload/{SN}
  - Example: /iot\_hub/upload/FC-8200H12345678
  - Messages sent by devices to servers will be published under this topic
- Will
  - When the device logs in, it will reserve the will on the MQTT server
  - The theme for publishing wills is device message publishing theme
  - The content of the will is in standard packet format, and the cmd command type is Will\_DeviceOffline
- Mask, the device supports using {SN} as a mask to replace with the actual SN of the device;

## Definition of Data Transmission Format

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## Message format

| Field                           | Type   | Explain                                                                                                                                                                                         |
|---------------------------------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| gzip Compression segment length | uint32 | Command type                                                                                                                                                                                    |
| gzip Length before compression  | uint32 | unix timestamp, representing the time when the command was issued                                                                                                                               |
| gzipCompression segment         | byte[] | The length of this segment is indicated by the 【gzip compressed segment length】<br>The JSON string in the gzip compressed segment, which is the original command body is encoded in UTF8 format |
| Additional file segments        | byte[] | Calculate the length of this segment from the message length -8- [gzip compression segment length]                                                                                              |

## Command body format

| Field   | Type   | Explain                                                                         |
|---------|--------|---------------------------------------------------------------------------------|
| Cmd     | string | Command type                                                                    |
| CmdTime | uint32 | unix timestamp, representing the time when the command was issued               |
| CmdID   | string | Command ID, generated by the generator, usually using auto increment ID or uuid |
| Body    | object | The main parameters of the command are defined by the command                   |

- Response message The message type of the response message is the main message type with an ACK ending, and the message that needs to be responded to must have a cmdid field

## Message Example

```
//1、Standard message structure
{
  "cmd": "UploadPeople", //Message type
  "CmdTime":123456789, //unix timestamp, representing the time when the command
was issued
  "CmdID":"6666", //Message id
  "body":{
    //Message parameters
  }
}
```

```
//2、Response message structure The message type of the response message is the
main message type with an ACK ending, and the message that needs to be responded
to must have a cmdid field
{
```

```
"cmd": "UploadPeopleACK", //Message type
"CmdTime":123456789, //unix timestamp, representing the time when the command
was issued
"CmdID":"6666", //Message id
"body":{
    //Message parameters
}
}
```

# Connect to keep-alive

## Device heartbeat protection package

### Brief description:

- After the device is turned on, it will periodically release secondary messages to allow the server to confirm that the device is currently online
- When the device triggers an alarm or the door status changes, this message will be sent to inform the server of the current status of the device

### Publisher

- Device

### Command

- KeepAlive

### Message parameters

| Field | Type   | Explain       |
|-------|--------|---------------|
| Cmd   | string | Command Type  |
| Body  | object | Device Status |

- Device status structure

| Field          | Type | Explain                                                                                            |
|----------------|------|----------------------------------------------------------------------------------------------------|
| RelayStatus    | int  | Relay state<br>0--indicates COM and NC are normally<br>1--indicates COM and NO are normally closed |
| KeepOpenStatus | int  | Normally open status<br>0--indicates normally closed<br>1--indicates normally open                 |

| Field            | Type   | Explain                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DoorSensorStatus | int    | Door magnetic status<br>0--indicates closed<br>1--indicates open                                                                                                                                                                                                                                                                                                                             |
| LockDoorStatus   | int    | Door locked status<br>0--indicates unlocked<br>1--indicates locked                                                                                                                                                                                                                                                                                                                           |
| AlarmStatus      | string | Door alarm status<br>An empty string indicates no alarm, otherwise there will be a specific alarm name<br>fire--fire alarm<br>blacklist--Blacklist alarm<br>anti--Anti disassembly alarm<br>illegal--Illegal verification<br>password--Duress alarm password<br>openTimeout--Opening timeout alarm<br>doorSensor--Door magnetic alarm<br>Separate multiple alarms with commas fire,blacklist |

## Message Example

```
{
  "cmd": "keepAlive",
  "CmdTime":123456789,//unix timestamp, representing the time when the command
was issued
  "body":{
    "RelayStatus": 0,
    "KeepOpenStatus": 0,
    "DoorSensorStatus": 0,
    "LockDoorStatus": 0,
    "AlarmStatus": ""
  }
}
```

# Offline device notification

## Device OfflineNotification

### Brief description:

- By utilizing the MQTT will function, in the event of an unexpected understanding disconnection on the device, the MQTT server will not send this reserved will

## Publisher

- Device

## Command

- Will\_DeviceOffline

## Message parameters

| Field | Type   | Explain      |
|-------|--------|--------------|
| Cmd   | string | Command type |

## Message Example

```
{
  "cmd": "Will_DeviceOffline"
}
```

# Device operating parameters

## Device basic Info SystemInfo

| Field              | Type   | Explain                                               |
|--------------------|--------|-------------------------------------------------------|
| DeviceSN           | string | Serial number                                         |
| DeviceName         | string | Device name                                           |
|                    |        |                                                       |
| FirmwareVersion    | string | Firmware version                                      |
| FingerprintVersion | string | Fingerprint algorithm version                         |
| FaceVersion        | string | Face algorithm version                                |
|                    |        |                                                       |
| Manufacturer       | string | Manufacturer                                          |
| ManufacturerPhone  | string | Manufacturer's phone number                           |
| Website            | string | Website                                               |
| ProductionDate     | string | Date of manufacture                                   |
| OEMText            | string | OEM custom text, can be filled in with 200 characters |
|                    |        |                                                       |

| Field           | Type   | Explain                                     |
|-----------------|--------|---------------------------------------------|
| AutoRestart     | int    | Daily automatic restart function switch     |
| AutoRestartTime | string | Daily automatic restart time, format HH: mm |

## Device working status Status

| Field            | Type   | Explain                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RunDays          | int    | System operation days                                                                                                                                                                                                                                                                                                                                                                        |
| FormatCount      | int    | Format times                                                                                                                                                                                                                                                                                                                                                                                 |
| WatchDogCount    | int    | Watchdog reset times                                                                                                                                                                                                                                                                                                                                                                         |
| BootTime         | int    | Startup time Unix timestamp (seconds)                                                                                                                                                                                                                                                                                                                                                        |
| RelayStatus      | int    | Relay state<br>0--indicates COM and NC are normally closed<br>1--indicates COM and NO are normally closed                                                                                                                                                                                                                                                                                    |
| KeepOpenStatus   | int    | Normally open<br>0-- indicates normally closed<br>1--indicates normally open                                                                                                                                                                                                                                                                                                                 |
| DoorSensorStatus | int    | Door magnetic status<br>0--Indicate off<br>1--Indicate on                                                                                                                                                                                                                                                                                                                                    |
| LockDoorStatus   | int    | Door locked state<br>0--Indicates unlocked<br>1--Indicates locked                                                                                                                                                                                                                                                                                                                            |
| AlarmStatus      | string | Door alarm status<br>An empty string indicates no alarm, otherwise there will be a specific alarm name<br>fire--Fire alarm<br>blacklist--Blacklist alarm<br>anti--Anti disassembly alarm<br>illegal--Illegal verification<br>password--Duress alarm password<br>openTimeout--Opening timeout alarm<br>doorSensor--Door magnetic alarm<br>Separate multiple alarms with commas fire,blacklist |

## Region and Language Language

| Field      | Type | Explain                                                                                                                                                                                                                                              |
|------------|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Language   | int  | Language 1 - Chinese;<br>2 - English; 3 - Traditional Chinese;4 - French;5 - Russian<br>6 - Portuguese;7 - Spanish;8 - Italian;9 - Japanese<br>10 - Korean;11 - Thai;12 - Arabic;13 - Portugal<br>14 - Turkey,15 - Indonesia,16 - Ukraine,17-Vietnam |
| SystemTime | int  | Device time Unix timestamp (seconds)                                                                                                                                                                                                                 |
| UseNTP     | int  | Enable NTP automatic time calibration 1--Enabled; 0--Disabled                                                                                                                                                                                        |
| TimeZone   | int  | Time zone of the device, value range:-12 - +14                                                                                                                                                                                                       |
| Volume     | int  | Volume size (range 0-10)                                                                                                                                                                                                                             |
| Voice      | int  | Voice playback switch 0,no broadcasting; 1,Broadcast                                                                                                                                                                                                 |

## human-computer interaction UI

| Field             | Type   | Explain                                                                                     |
|-------------------|--------|---------------------------------------------------------------------------------------------|
| DisplayBrightness | int    | Screen brightness setting 1-10                                                              |
| MenuPassword      | string | Menu password, only number, 4-8 digits or blank                                             |
| ShowIR            | int    | Display infrared images on the device 1--Enabled; 0--Disabled                               |
| ShowPersonPhoto   | int    | Display personnel profile picture after recognition 1--Enabled; 0--Disabled                 |
| PlayPersonName    | int    | Identify the name of the broadcaster after recognition 1--Enabled; 0--Disabled              |
| RecognitionButon  | int    | Before recognition, it is necessary to click the recognition button 1--Enabled; 0--Disabled |
| UnregisteredWarn  | int    | Unregistered personnel reminder 1--Enabled; 0--Disabled                                     |
| ShowPersonName    | int    | Whether to display personnel names after recognition 1--Enabled; 0--Disabled                |
| FillLight         | int    | Supplement light mode: 0: normally closed; 1: normally open; 2: automatic;                  |
| UseQRCode         | int    | QR code recognition switch 1--Enabled; 0--Disabled                                          |

## Data storage Storage

| Field             | Type  | Explain                                                                                                                                                                                                                                            |
|-------------------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RecordAutoCycle   | int   | Record full in cycle 1--Record full in cycle, 0--Record full not cycle, waiting for cleaning                                                                                                                                                       |
| SaveUnregistered  | int   | Save unregistered personnel records,0:Not stored,1:stored<br>Unregistered personnel refer to person who were not registered in the system during facial recognition , or whose card numbers were not registered in the system during register card |
| SaveRecordPicture | int   | Save on-site image 0,not save; 1,save                                                                                                                                                                                                              |
| PeopleStorageInfo | class | Personnel storage details                                                                                                                                                                                                                          |
| RecordStorageInfo | class | Record storage details                                                                                                                                                                                                                             |

- **Personnel storage information PeopleStorageInfo**

```
{
  "Person": {"Max": 5000, "Current":0 },//Personnel storage capacity   Max maximum capacity;   Current storage quantity
  "Face": {"Max": 5000, "Current":0 },//Face storage capacity
  "Card": {"Max": 5000, "Current":0 },//Card storage capacity
  "Fingerprint": {"Max": 5000, "Current":0 },//Fingerprint storage capacity
  "PalmVein": {"Max": 5000, "Current":0 },//Palm print storage capacity
  "Pasword": {"Max": 5000, "Current":0 },//Password storage capacity
  "Admin": {"Max": 5000, "Current":0 }//Administrator storage capacity
}
```

- **Record storage information RecordStorageInfo**

```
{
  "VerifyRecord": {"Max": 5000, "Current":0 },//Storage capacity for entry and exit records
  "DoorRecord": {"Max": 5000, "Current":0 },//Door magnetic recording stores information
  "SystemRecord": {"Max": 5000, "Current":0 },//System records storage capacity
  "RecordPhoto": {"Max": 5000, "Current":0 }//On site photo storage capacity
}
```

## Facial recognition Face

| Field           | Type | Explain                                                                                                           |
|-----------------|------|-------------------------------------------------------------------------------------------------------------------|
| FaceIR          | int  | Liveness detection,1 Open,0 Close                                                                                 |
| FaceIRThreshold | int  | Live detection threshold 1-99                                                                                     |
| FaceDistance    | int  | Identify distance 1--Short Distance (0.2-0.5m) ; 2--Middle Distance (0.2-1.5m) ; 3--Far Distance (0.2-1.5m above) |



| Field             | Type | Explain                                                                                                   |
|-------------------|------|-----------------------------------------------------------------------------------------------------------|
| FaceThreshold     | int  | The face recognition threshold is 1-99 The larger the face recognition threshold, the higher the accuracy |
| FPComparison      | int  | Fingerprint comparison threshold Value range: 1-100                                                       |
| FaceMask          | int  | Mask Detecting                                                                                            |
| FaceMaskThreshold | int  | Mask threshold 1-99 The higher the facial recognition threshold, the higher the accuracy                  |

## Temperature Detection BodyTemperature

| Field                     | Type  | Explain                                                                                                           |
|---------------------------|-------|-------------------------------------------------------------------------------------------------------------------|
| UseBodyTemperature        | Int   | Temperature measurement mode switch. 0: Non temperature mode 1: Temperature mode                                  |
| UseFahrenheitDisplay      | int   | Enable Fahrenheit temperature display,1:Enable,0:Disabled                                                         |
| TemperatureCompensate     | float | Temperature compensation value -10.0 -- +10.0                                                                     |
| TemperatureAlarmThreshold | float | Temperature alarm threshold Example 37.5                                                                          |
| TemperatureDisplay        | int   | Whether to display body temperature ` 0--Prohibit displaying body temperature; 1--Display temperature information |

## Server parameters NetworkServer

| Field                 | Type   | Explain                                                     |
|-----------------------|--------|-------------------------------------------------------------|
| UseTCPClient          | Int    | Use TCPClient to connect to server 1-- Enable; 0--Disabled; |
| UseUDPClient          | Int    | Use UDPClient to connect to server 1-- Enable; 0--Disabled; |
| ServerAddress         | string | Server Address tcp or udp Protocol Server Address           |
| ServerPort            | int    | Server port number                                          |
| KeepaliveTime         | int    | Keepalive package interval time 1-65535 s                   |
|                       |        |                                                             |
| UseHTTPClient         | int    | Enable HTTPClient protocol 1--Enable; 0-- Disabled;         |
| HTTPClient_ServerAddr | string | HTTP protocol server address                                |

| Field                         | Type   | Explain                                                                                           |
|-------------------------------|--------|---------------------------------------------------------------------------------------------------|
| HTTPClient_KeepaliveTime      | int    | Time interval for keep alive packets in HTTP protocol                                             |
| HTTPClient_UseGZIP            | int    | HTTP Is GZIP compression used when making HTTP protocol requests? 0-- Not used; 1--Use            |
| HTTPClient_ProtocolType       | int    | Protocol type of HTTPClient<br>100 --- HTTPv1<br>200 ---HTTPv2                                    |
|                               |        |                                                                                                   |
| UseMQTTClient                 | int    | Whether to start MQTTClient protocol 1-- Enable; 0--Disabled;                                     |
| UseMQTTSSL                    | int    | Enable MQTT SSL Secure Socket 1--Enable; 0--Disabled;                                             |
| MQTTServerAddr                | string | MQTT server address <a href="http://www.abc.com">www.abc.com</a>                                  |
| MQTTPort                      | int    | MQTT server port number                                                                           |
| MQTTLoginName                 | string | Login username in MQTT protocol                                                                   |
| MQTTLoginPassword             | string | Login password in MQTT protocol                                                                   |
| MQTTPublishTopic              | string | Topics used by devices to send data in MQTT protocol                                              |
| MQTTSubscribeTopic            | string | Topics that devices in the MQTT protocol need to subscribe to when receiving data                 |
| MQTT_KeepaliveTime            | int    | The keep alive packet interval time of MQTT protocol                                              |
| MQTT_UseGZIP                  | int    | Does MQTT use GZIP compression 0-- Not used; 1--Use                                               |
|                               |        |                                                                                                   |
| UseWebSocketClient            | int    | Whether to enable WebSocketClient protocol 1--Enable; 0--Disabled;                                |
| WebSocketClient_ServerAddr    | string | WebSocket protocol Server address<br>ws://192.168.1.1/websocket or<br>wss://192.168.1.1/websocket |
| WebSocketClient_KeepaliveTime | int    | The keep-alive packet interval time of WebSocket Client protocol                                  |
| WebSocketClient_UseGZIP       | int    | WebSocket Whether to use GZIP compression 0--Not use; 1--Use                                      |
| WebSocketClient_ProtocolType  | int    | Protocol types of WebSocket                                                                       |

| Field   | Type   | Explain                                                                              |
|---------|--------|--------------------------------------------------------------------------------------|
|         |        |                                                                                      |
| UseYZW  | int    | Whether to enable the HTTPClient protocol of Yunzhu platform 1--Enable; 0--Disabled; |
| YZWAddr | string | Yunzhu platform protocol Server Address                                              |

## Device network parameters Network

| Field          | Type   | Explain                                         |
|----------------|--------|-------------------------------------------------|
| UseWired       | int    | Wired network switch,1:On,0:Off                 |
| WiredDHCP      | int    | Wired network automatic ip,1:On,0:Off           |
| WiredIP        | string | Wired network IP address("///192.168.0.110")    |
| WiredIPMask    | string | Wired network subnet mask("///255.255.255.0")   |
| WiredGteway    | string | Wired network gateway("///192.168.0.1")         |
| WiredDNS       | string | Dns("///192.168.0.1")                           |
| WiredMAC       | string | Wired network MAC address                       |
|                |        |                                                 |
| UseWIFI        | int    | Wireless network switch,1:On,0:Off              |
| WIFIAPName     | string | Wireless network account                        |
| WIFIAPPassword | string | Wireless network password                       |
| WIFIMAC        | string | Wireless network MAC address                    |
| WIFIDHCP       | int    | Wireless network automatic ip,1:On,0:Off        |
| WIFIIP         | string | Wireless network IP address("///192.168.0.110") |
| WIFIIPMask     | string | Wireless network subnet mask"//255.255.255.0")  |
| WIFIGteway     | string | Wireless network gateway("///192.168.0.1")      |
| WIFIDNS        | string | Wireless Network Dns("///192.168.0.1")          |
|                |        |                                                 |
|                |        |                                                 |
| UseWebPage     | int    | Web page management switch,1:On,0:Off           |
| HTTPPort       | int    | Web management page port number, 1-65534        |
| HTTPSPort      | int    | Web management page port number, 1-65534        |

| Field           | Type   | Explain                                                                       |
|-----------------|--------|-------------------------------------------------------------------------------|
| WebPageUseSSL   | int    | Enable SSL on device web page `` SSL certificate is self signed using OpenSSL |
|                 |        |                                                                               |
| UseUDP          | int    | UDP port switch ,1:On,0:Off                                                   |
| UDPPort         | int    | UDP port number (used by UDP protocol)                                        |
| ConnectPassword | string | UDP protocol communication password 32 characters                             |
|                 |        |                                                                               |
| UseTelnet       | int    | Linux Telnet function switch,1:On, 0:Off                                      |
| TelnetPort      | int    | Telnet Port No.                                                               |
|                 |        |                                                                               |
| UseRTSP         | int    | RTSP video streaming ,1:On,0:Off                                              |
| RTSPPort        | int    | RTSP Port No.                                                                 |
| RTSPUser        | string | RTSP User Name                                                                |
| RTSPPassword    | string | RTSP Password                                                                 |

## Access control parameters Door

| Field                   | Type | Required | Explain                                                     |
|-------------------------|------|----------|-------------------------------------------------------------|
| CardBytes               | int  | Yes      | Card number byte; 3、4、8; 0-- indicates disable card reading |
| AccessType              | int  | Yes      | Entry and exit type 0,entry; 1,exit                         |
| WgFormat                | int  | Yes      | WG output format 26 / 34/66                                 |
| WGContent               | int  | Yes      | WG output content: 1--User ID; 2-- Card number              |
| ReleaseTime             | int  | Yes      | Opening Holding Time 0-65535 (s) 。 0 represents 0.5s        |
| DelayOpenDoorTime       | int  | Yes      | Delay unlocking time 0-65535 (s) 。 0 means disabled         |
| FreeOpen                | int  | Yes      | Opening without verification 1-- Enable; 0--Diasbled        |
| OpenInterval            | int  | Yes      | Repetitive recognition interval 0-- Disabled ; 1-65535 (ms) |
| OpenInterval_SaveRecord | int  | Yes      | Repeat interval record storage setting 0,Not save; 1,Save   |

| Field             | Type   | Required | Explain                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------------|--------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Relay             | int    | Yes      | Whether the relay support bistability?<br>` ` 1 is supported,0 is not supported                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| ShortMessage      | string | Yes      | Short messages after legally verification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| VerificationType  | int    | Yes      | Verification method;<br>1、 Standard mode; 2、 Face/fingerprint/palm vein/card+password;<br>2、 Face/fingerprint/palm vein/card+password;<br>3、 Card+face/fingerprint/palm vein/password;<br>4、 Multi person attendance 5、 People and Identification card comparison;<br>6、 Card+face/fingerprint/palm vein+password;<br>7、 Card+fingerprint/palm vein+face recognition;<br>8、 Fingerprint/palm vein+face+password;<br>9、 Fingerprint+palm vein+face;<br>10、 Palm vein+face; 11、 Fingerprint+face;<br>12、 Only use the palm vein; 13、 Only use fingerprints; 14、 Only use card; 15、 Only use password;<br>16、 People and Identification card comparison: Open door+auto registered (Identification card+face+auto registered);; |
| OverdueRemind     | int    | Yes      | Permission expiration prompt 1-- Enable; 0--Disabled                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| OverdueRemind_Day | int    | Yes      | Permission expiration prompt Validity threshold, minimum remaining valid days. If the number of days is below this threshold, it will prompt that the validity period is about to expire, value range: 1-255. 0--indicates closed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                   |        |          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| TimingOpen        | int    | Yes      | Timing Normally Opened Function 1-- Enable; 0--Diabled                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

| Field                  | Type   | Required | Explain                                                                                                                                                                                                                                                                                                                                                                                             |
|------------------------|--------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TimingOpen_mode        | int    | Yes      | Timing Normally Opened. Automatic on mode: ``1、 After passing the legal authentication, it can be normally open within the specified period of time. 2. Those marked as normally open privilege in the authorization can be normally open after passing the authentication within the specified period of time. 3. Automatic switch, the door will automatically open and close when the time is up |
| TimingOpen_timegroup   | object | Yes      | Timing Normally Opened. Normally opened time zone Use weekly time zone structure                                                                                                                                                                                                                                                                                                                    |
|                        |        |          |                                                                                                                                                                                                                                                                                                                                                                                                     |
| TimingLocked           | int    | Yes      | Timing Locked Function 1--Enable; 0--Diabled                                                                                                                                                                                                                                                                                                                                                        |
| TimingLocked_timegroup | object | Yes      | Timing Locked. Locking Time Zone Use weekly time zone structure                                                                                                                                                                                                                                                                                                                                     |
|                        |        |          |                                                                                                                                                                                                                                                                                                                                                                                                     |
| VisitorRootPassword    | string | Yes      | Visitor root password                                                                                                                                                                                                                                                                                                                                                                               |
| MultiPerson            | int    | Yes      | Multi person combination opening the door, number of people; 1-50;                                                                                                                                                                                                                                                                                                                                  |

- VerificationType Verification method

- 1、 Standard mode default value
- 2、 Face/fingerprint/palm vein/card+password
- 3、 card+face/fingerprint/palm vein/password
- 4、 Multi person attendance
- 5、 People and Identification Card Comparison
- 6、 Card+face/fingerprint/palm vein+password
- 7、 Card+fingerprint/palm vein+face recognition
- 8、 Fingerprint/palm vein+face+password
- 9、 Fingerprint+palm vein+face
- 10、 Palm vein+face
- 11、 Fingerprint+Face
- 12、 Only Palm vein
- 13、 Only Fingerprint
- 14、 Only Card
- 15、 Only Password
- 16、 People and Identification Card Comparison + Auto Registered  
(Identification Card + Face + Auto Registered)

- **Weekly time zone format**

```
{
    week1: "00:00-23:59", //Mon
    week2: "00:00-23:59",
    week3: "00:00-23:59",
    week4: "00:00-23:59",
    week5: "00:00-23:59",
    week6: "00:00-23:59",
    week7: "00:00-23:59" //Sun
}
```

- Week1 Indicates Monday
- Week2 Indicates Tuesday
- Week3 Indicates Wednesday
- Week4 Indicates Thursday
- Week5 Indicates Friday
- Week6 Indicates Saturday
- Week7 Indicates Sunday
- Each week field represents the time zone setting for a day
- You can set 8 sub time zone in a day with the format of start time end time/start time-end time/.....

"01:00-01:59/02:00-02:59/03:00-03:59/04:00-04:59/05:00-05:59/06:00-06:59/07:00-07:59/08:00-08:59"

The above string defines 8 sub time zone in a day. A maximum of 8 sub time zone can be defined per day

"01:00-01:59/02:00--02:59"

The above string defines 2 sub time zone in a day, and the other 6 time zone are empty and invalid.

- If the day of the week is empty and no time zone is set, this field can be omitted

```
{
    //At this time, only week 1 and week 7 are defined, and other time
    zone are empty
    week1: "01:00-02:00",
    week7: "03:00-04:00"
}
```

## Elevator functional parameters Elevator

| Field         | Type            | Required | Explain                                              |
|---------------|-----------------|----------|------------------------------------------------------|
| UseElevator   | int             | Yes      | Elevator function switch, 1: on, 0: off              |
| ElevatorPorts | [] Object array | Yes      | ElevatorPort object array defines elevator port list |

## Elevator Port Object

| Field       | Type | Required | Explain                                                                                                                                                |
|-------------|------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Num         | int  | Yes      | Elevator port number 1-64                                                                                                                              |
| RelayType   | int  | Yes      | Elevator relay (COM&NO normally closed, COM&NC normally closed) <br> Value range: 1、COM&NC normally closed (default value)) ; 2、COM&NO normally closed |
| ReleaseTime | int  | Yes      | The maximum output unlocking time is 65535 seconds. 0 represents 0.5 seconds                                                                           |
| TimingOpen  | obj  | Yes      | Timing normally open function structure                                                                                                                |

- timingOpen Timing normally open function structure

| Field     | Type | Required | Explain                                                                                                                                                                                                                                                                                                        |
|-----------|------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use       | int  | Yes      | Function switch,0--Diabled; 1--Enabled                                                                                                                                                                                                                                                                         |
| Open      | int  | Yes      | Automatic opening mode: <br> 1、1. After passing the legal authentication, the door can be normally opened within the specified time zone <br> 2、 If the authorization is marked as normally open privilege and authenticated within the specified time zone, then it can be normally open <br> 3、自动开关,到时间自动开关门 |
| Timegroup | obj  | Yes      | Normally open time zone, using weekly time zone structure                                                                                                                                                                                                                                                      |

## Alarm parameters Alarm

| Field             | Type | Required | Explain                                                                                                  |
|-------------------|------|----------|----------------------------------------------------------------------------------------------------------|
| FireAlarm         | int  | Yes      | Fire alarm,0,off; 1,on                                                                                   |
|                   |      |          |                                                                                                          |
| DoorLongOpenAlarm | int  | Yes      | Open door timeout alarm switch,1:on,0:off                                                                |
| DoorLongOpenTime  | int  | Yes      | Open door timeout, if the door is opened for more than this time, an alarm will be triggered 1-65535 (s) |
|                   |      |          |                                                                                                          |
| DoorSensorAlarm   | int  | Yes      | Door magnetic alarm,0,off; 1,on                                                                          |



| Field                         | Type   | Required | Explain                                                                                                                                                                            |
|-------------------------------|--------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DoorSensorAlarmTimegroup      | class  | Yes      | Non alarm time zone of door magnetic alarm, weekly time zone format                                                                                                                |
|                               |        |          |                                                                                                                                                                                    |
| BlacklistAlarm                | int    | Yes      | Blacklist alarm,0,off; 1,on                                                                                                                                                        |
|                               |        |          |                                                                                                                                                                                    |
| AntiDisassemblyAlarm          | int    | Yes      | Anti disassembly alarm function switch,0,off; 1,on                                                                                                                                 |
|                               |        |          |                                                                                                                                                                                    |
| IllegalVerificationAlarm      | int    | Yes      | Illegal verification alarm function,0,off; 1,on                                                                                                                                    |
| IllegalVerificationAlarmLimit | int    | Yes      | Illegal verification alarm function - number of illegal verifications,1-255                                                                                                        |
|                               |        |          |                                                                                                                                                                                    |
| UseUserCloseAlarm             | int    | Yes      | Allow users to verify the alarm release switch switch,0,off; 1,on                                                                                                                  |
|                               |        |          |                                                                                                                                                                                    |
| PasswordAlarm                 | int    | Yes      | Duress alarm password function,0,off; 1,on                                                                                                                                         |
| PasswordAlarm_Password        | string | Yes      | Duress alarm password,input this password will trigger an alarm. Passwords only support numbers and can contain 0.                                                                 |
| PasswordAlarm_Mode            | string | Yes      | The working mode when the duress alarm occurs 1--Do not open the door, alarm output 2--Open the door, alarm output ``3--Lock the door, alarm, can only be unlocked by software```` |

## Device opening time zone Timegroup

| Field      | Type            | Required | Explain                  |
|------------|-----------------|----------|--------------------------|
| TimeGroups | [] Object array | Yes      | Device opening time zone |

**Format description of device opening time zone**

```
//Parameters
TimeGroups:[
  {
    Num:1,
    week1:"00:00-23:59", //Monday
    week2:"00:00-23:59",
    week3:"00:00-23:59",
    week4:"00:00-23:59",
    week5:"00:00-23:59",
    week6:"00:00-23:59",
    week7:"00:00-23:59" //Sunday
  },
  {
    Num:2,
    week1:"01:00-01:59/02:00-02:59/01:00-01:59/02:00-02:59",
    ..
    week7:"00:00-00:00"
  },
  ..
  {
    Num:64,
    week1:"00:00-00:00",
    ..
    week7:"00:00-00:00"
  }
]
```

- The maximum num during the opening time zone of the device is 64, indicating that the device has a maximum of 64 sets of opening time zone
- week1 Indicates Monday
- week2 Indicates Tuesday
- week3 Indicates Wednesday
- week4 Indicates Thursday
- week5 Indicates Friday
- week6 Indicates Saturday
- week7 Indicates Sunday
- Each week field represents the time zone setting for a day
- You can set 8 sub time zone in a day with the format of start time-end time/start time-end time/.....

```
"01:00-01:59/02:00-02:59/03:00-03:59/04:00-04:59/05:00-05:59/06:00-06:59/07:00-07:59/08:00-08:59"
```

The above string defines 8 sub time zone in a day. A maximum of 8 sub time zone can be defined per day

```
"01:00-01:59/02:00-02:59"
```

The above string defines 2 sub time zone in a day, and the other 6 time zone are empty and invalid.

- If the day of the week is empty and no time zone is set, this field can be omitted

```
{
    Num:64, //At this time, only week 1 and week 7 are defined, and
    other time zone are empty
    week1:"01:00-02:00",
    week7:"03:00-04:00"
}
```

## Holiday of Device Holiday

| Field    | Type | Required | Explain |
|----------|------|----------|---------|
| Holidays | []   | Yes      | Holiday |

```
[
  {"Num":1,"Date":"2020-10-01","Type":1,"Cycle":1},
  {"Num":2,"Date":"2020-10-02","Type":2,"Cycle":0},
  ...
]
```

Use object array format during holidays, with each object containing two fields, num and date.

The device currently supports 360 sets of holidays

On holidays, it is prohibited to open doors (can set the permission for holiday passage)

### Holiday Object Field Description

| Field | Type   | Required | Explain                                                                                                   |
|-------|--------|----------|-----------------------------------------------------------------------------------------------------------|
| Num   | int    | Yes      | Number of holidays, used when binding personnel permissions                                               |
| Date  | string | Yes      | Holiday Date Year-Month-Day Example: 2020-10-01                                                           |
| Type  | int    | No       | Holiday control range,1-- All day; 2--Morning 00:00-12:00; 3--Afternoon(12:00-23:59),default value is 1 ; |
| Cycle | int    | No       | Whether to cycle annually,1--Cycle annually; 0--Non cycle; Default value is 0;                            |

## Alarm clock AlarmClock

### Up to 24 alarms

| Field       | Type | Required | Explain     |
|-------------|------|----------|-------------|
| AlarmClocks | []   | Yes      | Alarm Clock |

```
[
  {"Num":1,"Clock":"12:00","Times":10},
  {"Num":2,"Clock":"13:00","Times":10},
  {"Num":3,"Clock":"14:00","Times":10},
  ...
]
```

The alarm clock uses an object array format, with each object containing three fields, Num 、 Clock 、 Times.

The device currently supports 24 sets of alarms

### Field Description of Alarm Clock Object

| Field | Type | Required | Explain                                                   |
|-------|------|----------|-----------------------------------------------------------|
| Num   | int  | Yes      | The serial number of the alarm clock, value range is 1-24 |
| Clock | int  | Yes      | Alarm time HHmmformat example: 1230 means 12:30 pm        |
| Times | int  | No       | Alarm ringing time ,unit seconds value range 0-255        |

## Device Function List FunctionList

```
{
  "FunctionList" :{
    //Temperature detection
    "BodyTemperature": true,
    //Fingerprint
    "Fingerprint": true,
    //Palm Print
    "Palmvein": true,
    //Face Recognition
    "Face": true,
    //QR Code
    "QRCode": true,
    //Mask Detecting
    "FaceMask": true,
    //Helmet Detecting
    "safetyHelmet": true,
    //Elevator
    "Lift": true,
    //Alarm Clock
    "AlarmClock": true,
    //excel Import and Export
    "ExcelFile": true,
    //zip Import
    "ZipFile": true,
    //Number of time zone
    "TimeGreoup": true,
    //Wireless network
    "WIFI": true,
    //HTTPClient v1
    "HTTPClient_V1": true,
```

```

    //HttpClient v2
    "HttpClient_v2": true,
    //MQTT
    "MQTT": true,
    //Yunzhu Platform
    "YZW": true,
    //websocket v1
    "websocket_v1": true,
    //websocket v2
    "websocket_v2": true
  }
}

```

## The device actively uploads working parameters

### Brief description:

- Send once when the device is turned on
- When the device properties change, they will be uploaded again
- When the server requests to read the working parameters of the device, the device will also send this command to upload the working parameters to the server

### Publisher

- Device

### Command

- UploadWorkSetting

### Message parameters

| Field | Type   | Explain                                                                                      |
|-------|--------|----------------------------------------------------------------------------------------------|
| Cmd   | string | Command type                                                                                 |
| CmdID | string | Command ID, the ID in the command issued by the server (if the server requires transmission) |
| body  | object | Device operating parameters                                                                  |

### Message Example

```

{
  "cmd": "UploadWorkSetting",
  "CmdID": "123456",
  "CmdTime": 123456789, //Unix timestamp, representing the time of the
  command was issued
  "body":

```

```
{
  "FireAlarm": 0,
  "DoorLongOpenAlarm": 0,
  "DoorLongOpenTime": 0,
  "DoorSensorAlarm": 0,
  "DoorSensorAlarmTimegroup": {
    "week1": "",
    "week2": "",
    "week3": "",
    "week4": "",
    "week5": "",
    "week6": "",
    "week7": ""
  },
  "BlacklistAlarm": 1,
  "AntiDisassemblyAlarm": 1,
  "IllegalVerificationAlarm": 0,
  "IllegalVerificationAlarmLimit": 30,
  "UseUserCloseAlarm": 1,
  "PasswordAlarm": 0,
  "PasswordAlarm_Password": "",
  "PasswordAlarm_Mode": 1,
  "AlarmClocks": [
    {
      "Num": 1,
      "Clock": "00:00",
      "Times": 0
    },
    {
      "Num": 24,
      "Clock": "00:00",
      "Times": 0
    }
  ],
  "UseBodyTemperature": 1,
  "UseFahrenheitDisplay": 1,
  "TemperatureCompensate": 0,
  "TemperatureAlarmThreshold": 37.29999,
  "TemperatureDisplay": 1,
  "CardBytes": 3,
  "AccessType": 0,
  "WgFormat": 26,
  "WGContent": 1,
  "ReleaseTime": 3,
  "FreeOpen": 0,
  "OpenInterval": 2,
  "OpenInterval_SaveRecord": 0,
  "Relay": 0,
  "ShortMessage": "",
  "VerificationType": 1,
  "OverdueRemind": 1,
  "OverdueRemind_Day": 3,
  "TimingOpen": 0,
  "TimingOpen_mode": 1,
  "TimingOpen_timegroup": {
```

```

        "week1": "",
        "week2": "",
        "week3": "",
        "week4": "",
        "week5": "",
        "week6": "",
        "week7": ""
    },
    "TimingLocked": 0,
    "TimingLocked_timegroup": {
        "week1": "",
        "week2": "",
        "week3": "",
        "week4": "",
        "week5": "",
        "week6": "",
        "week7": ""
    },
    "VisitorRootPassword": "",
    "MultiPerson": 1,
    "UseElevator": 0,
    "ElevatorPorts": [
        {
            "Num": 1,
            "RelayType": 2,
            "ReleaseTime": 2,
            "TimingOpen": {
                "Use": 0,
                "Open": 3,
                "Timegroup": {
                    "week1": "",
                    "week2": "",
                    "week3": "",
                    "week4": "",
                    "week5": "",
                    "week6": "",
                    "week7": ""
                }
            }
        },
        {
            "Num": 64,
            "RelayType": 2,
            "ReleaseTime": 2,
            "TimingOpen": {
                "Use": 0,
                "Open": 3,
                "Timegroup": {
                    "week1": "",
                    "week7": ""
                }
            }
        }
    ],
    "FaceIR": 1,

```

```
"FaceIRThreshold": 5,
"FaceDistance": 3,
"FaceThreshold": 58,
"FPComparison": 80,
"FaceMask": 0,
"FaceMaskThreshold": 65,
"Holidays": [],
"Language": 1,
"SystemTime": 1718421632,
"UseNTP": 1,
"TimeZone": 8,
"Volume": 6,
"Voice": 1,
"ConnectPassword": "",
"UseWired": 1,
"WiredDHCP": 0,
"WiredIP": "192.168.1.103",
"WiredIPMask": "255.255.255.0",
"WiredGteway": "192.168.1.1",
"WiredDNS": "192.168.1.1",
"WiredMAC": "7E-87-16-C1-E6-51",
"UseWiFi": 0,
"WIFIDHCP": 0,
"WIFIIP": "192.168.1.150",
"WIFIIPMask": "255.255.255.0",
"WIFIGteway": "192.168.1.1",
"WIFIDNS": "192.168.1.1",
"WIFIMAC": "34-7D-E4-2D-63-3B",
"WIFIAPName": "",
"WIFIAPPassword": "",
"UseWebPage": 1,
"HTTPPort": 80,
"HTTPSPort": 443,
"WebPageUseSSL": 1,
"UseUDP": 1,
"UDPPort": 8101,
"UseTelnet": 1,
"TelnetPort": 23,
"UseRTSP": 1,
"RTSPPort": 554,
"RTSPUser": "admin",
"RTSPPassword": "12345678",
"UseTCPClient": 0,
"UseUDPClient": 0,
"ServerAddress": "47.92.31.75",
"ServerPort": 9003,
"KeepaliveTime": 30,
"UseHTTPClient": 0,
"HTTPClient_ProtocolType": 100,
"HTTPClient_ServerAddr": "http://192.168.1.100",
"HTTPClient_KeepaliveTime": 30,
"HTTPClient_UseGZIP": 0,
"UseMQTTClient": 0,
"UseMQTTSSL": 0,
"MQTTServerAddr": "192.168.1.100",
```



```
"MQTTPort": 0,
"MQTTLoginName": "",
"MQTTLoginPassword": "",
"MQTTPublishTopic": "",
"MQTTSubscribeTopic": "",
"MQTT_KeepaliveTime": 30,
"MQTT_UseGZIP": 0,
"UseWebSocketClient": 0,
"WebSocketClient_ProtocolType": 100,
"WebSocketClient_ServerAddr": "ws://192.168.1.100/ws",
"WebSocketClient_UseGZIP": 0,
"WebSocketClient_KeepaliveTime": 30,
"UseYZW": 0,
"YZWAddr": "http://192.168.1.10",
"RunDays": 0,
"FormatCount": 0,
"WatchDogCount": 7,
"BootTime": 1718418194,
"RelayStatus": 0,
"KeepOpenStatus": 0,
"DoorSensorStatus": 0,
"LockDoorStatus": 0,
"AlarmStatus": "\\\"\\\"",
"RecordAutoCycle": 0,
"SaveUnregistered": 1,
"SaveRecordPicture": 1,
"PeopleStorageInfo": {
  "Person": {
    "Max": 20000,
    "Current": 1
  },
  "Face": {
    "Max": 20000,
    "Current": 1
  },
  "Card": {
    "Max": 20000,
    "Current": 0
  },
  "Fingerprint": {
    "Max": 0,
    "Current": 0
  },
  "PalmVein": {
    "Max": 10000,
    "Current": 0
  },
  "Pasword": {
    "Max": 20000,
    "Current": 0
  },
  "Admin": {
    "Max": 5,
    "Current": 0
  }
}
```

```
    },
    "RecordStorageInfo": {
      "VerifyRecord": {
        "Max": 1000000,
        "Current": 4
      },
      "DoorRecord": {
        "Max": 10000,
        "Current": 4
      },
      "SystemRecord": {
        "Max": 10000,
        "Current": 7
      },
      "RecordPhoto": {
        "Max": 10000,
        "Current": 4
      }
    },
    "DevicesN": "FC-8190H24052799",
    "DeviceName": "",
    "FirmwareVersion": "8.46",
    "FingerprintVersion": "-",
    "FaceVersion": "6.01",
    "Manufacturer": "",
    "ManufacturerPhone": "",
    "Website": "",
    "ProductionDate": "2024-06-15",
    "OEMText": "",
    "AutoRestart": 0,
    "AutoRestartTime": "02:00",
    "TimeGroups": [
      {
        "Num": 1,
        "Week1": "00:00-23:59",
        "Week2": "00:00-23:59",
        "Week3": "00:00-23:59",
        "Week4": "00:00-23:59",
        "Week5": "00:00-23:59",
        "Week6": "00:00-23:59",
        "Week7": "00:00-23:59"
      },
      {
        "Num": 64,
        "Week1": "",
        "Week7": ""
      }
    ],
    "DisplayBrightness": 6,
    "MenuPassword": "0000",
    "ShowIR": 0,
    "ShowPersonPhoto": 1,
    "PlayPersonName": 1,
    "RecognitionButon": 0,
    "UnregisteredWarn": 0,
```

```
"ShowPersonName": 1,
"FillLight": 2,
"FunctionList": {
  "BodyTemperature": true,
  "Fingerprint": false,
  "Palmvein": true,
  "Face": true,
  "QRCode": true,
  "FaceMask": true,
  "SafetyHelmet": false,
  "Lift": true,
  "AlarmClock": true,
  "ExcelFile": false,
  "ZipFile": false,
  "TimeGreoup": 64,
  "WIFI": true,
  "HTTPClient_V1": true,
  "HTTPClient_V2": true,
  "MQTT": true,
  "YZW": true,
  "websocket_V1": true,
  "websocket_V2": true
}
}
```

## The server requires the device to upload working parameters

### Brief description:

- When the server needs to view the working parameters stored inside the device, the server sends this command to request the device to send the \* \* UploadWorkSetting \* \* command to upload the working parameters to the server

### Publisher

- Server

### Command

- ReadWorkSetting

### Message parameters

| Field | Type   | Explain                                                |
|-------|--------|--------------------------------------------------------|
| Cmd   | string | Command type                                           |
| CmdID | string | Command ID, the ID in the command issued by the server |

## Message Example

```
{
  "cmd": "ReadWorkSetting",
  "CmdTime":123456789,//Unix timestamp, representing the time of the command
  was issued
  "CmdID":"123456"
}
```

## Server issues work parameters

### Brief description:

- When the working parameters of the device saved on the server need to be changed and sent to the device, the server will actively publish a message

### Publisher

- Server

### Command

- PushWorkSetting

### Message parameters

| Field | Type   | Explain                                                                                                                                                                |
|-------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cmd   | string | Command type                                                                                                                                                           |
| CmdID | string | Command ID, when the device returns the corresponding command response packet, it will carry . It is usually generated using methods such as self increment ID or uuid |
| Body  | object | Parameter fields that need to be set                                                                                                                                   |

## Message Example

```
{
  "cmd": "PushWorkSetting",
  "CmdTime":123456789,//Unix timestamp, representing the time of the
  command was issued
  "CmdID": "123456",
  "body":
  {
    "FireAlarm": 0,
    "DoorLongOpenAlarm": 0,
    "DoorLongOpenTime": 0,
    "DoorSensorAlarm": 0,
  }
}
```

```
"DoorSensorAlarmTimegroup": {
    "Week1": "",
    "Week2": "",
    "Week3": "",
    "Week4": "",
    "Week5": "",
    "Week6": "",
    "Week7": ""
},
"BlacklistAlarm": 1,
"AntiDisassemblyAlarm": 1,
"IllegalVerificationAlarm": 0,
"IllegalVerificationAlarmLimit": 30,
"UseUserCloseAlarm": 1,
"PasswordAlarm": 0,
"PasswordAlarm_Password": "",
"PasswordAlarm_Mode": 1,
"AlarmClocks": [
    {
        "Num": 1,
        "Clock": "00:00",
        "Times": 0
    },
    {
        "Num": 24,
        "Clock": "00:00",
        "Times": 0
    }
],
"UseBodyTemperature": 1,
"UseFahrenheitDisplay": 1,
"TemperatureCompensate": 0,
"TemperatureAlarmThreshold": 37.29999,
"TemperatureDisplay": 1,
"CardBytes": 3,
"AccessType": 0,
"WgFormat": 26,
"WGContent": 1,
"ReleaseTime": 3,
"FreeOpen": 0,
"OpenInterval": 2,
"OpenInterval_SaveRecord": 0,
"Relay": 0,
"ShortMessage": "",
"VerificationType": 1,
"OverdueRemind": 1,
"OverdueRemind_Day": 3,
"TimingOpen": 0,
"TimingOpen_mode": 1,
"TimingOpen_timegroup": {
    "Week1": "",
    "Week2": "",
    "Week3": "",
    "Week4": "",
    "Week5": "",
```

```

        "week6": "",
        "week7": ""
    },
    "TimingLocked": 0,
    "TimingLocked_timegroup": {
        "week1": "",
        "week2": "",
        "week3": "",
        "week4": "",
        "week5": "",
        "week6": "",
        "week7": ""
    },
    "VisitorRootPassword": "",
    "MultiPerson": 1,
    "UseElevator": 0,
    "ElevatorPorts": [
        {
            "Num": 1,
            "RelayType": 2,
            "ReleaseTime": 2,
            "TimingOpen": {
                "Use": 0,
                "Open": 3,
                "Timegroup": {
                    "week1": "",
                    "week2": "",
                    "week3": "",
                    "week4": "",
                    "week5": "",
                    "week6": "",
                    "week7": ""
                }
            }
        },
        {
            "Num": 64,
            "RelayType": 2,
            "ReleaseTime": 2,
            "TimingOpen": {
                "Use": 0,
                "Open": 3,
                "Timegroup": {
                    "week1": "",
                    "week7": ""
                }
            }
        }
    ],
    "FaceIR": 1,
    "FaceIRThreshold": 5,
    "FaceDistance": 3,
    "FaceThreshold": 58,
    "FPComparison": 80,
    "FaceMask": 0,

```

```
"FaceMaskThreshold": 65,  
"Holidays": [],  
"Language": 1,  
"SystemTime": 1718421632,  
"UseNTP": 1,  
"TimeZone": 8,  
"Volume": 6,  
"Voice": 1,  
"ConnectPassword": "",  
"UseWired": 1,  
"WiredDHCP": 0,  
"WiredIP": "192.168.1.103",  
"WiredIPMask": "255.255.255.0",  
"WiredGteway": "192.168.1.1",  
"WiredDNS": "192.168.1.1",  
"WiredMAC": "7E-87-16-C1-E6-51",  
"UseWiFi": 0,  
"WIFIDHCP": 0,  
"WIFIIP": "192.168.1.150",  
"WIFIIPMask": "255.255.255.0",  
"WIFIGteway": "192.168.1.1",  
"WIFIDNS": "192.168.1.1",  
"WIFIMAC": "34-7D-E4-2D-63-3B",  
"WIFIAPName": "",  
"WIFIAPPassword": "",  
"UseWebPage": 1,  
"HTTPPort": 80,  
"HTTPSPort": 443,  
"WebPageUseSSL": 1,  
"UseUDP": 1,  
"UDPPort": 8101,  
"UseTelnet": 1,  
"TelnetPort": 23,  
"UseRTSP": 1,  
"RTSPPort": 554,  
"RTSPUser": "admin",  
"RTSPPassword": "12345678",  
"UseTCPClient": 0,  
"UseUDPClient": 0,  
"ServerAddress": "47.92.31.75",  
"ServerPort": 9003,  
"KeepaliveTime": 30,  
"UseHTTPClient": 0,  
"HTTPClient_ProtocolType": 100,  
"HTTPClient_ServerAddr": "http://192.168.1.100",  
"HTTPClient_KeepaliveTime": 30,  
"HTTPClient_UseGZIP": 0,  
"UseMQTTClient": 0,  
"UseMQTTSSL": 0,  
"MQTTServerAddr": "192.168.1.100",  
"MQTTPort": 0,  
"MQTTLoginName": "",  
"MQTTLoginPassword": "",  
"MQTTPublishTopic": "",  
"MQTTSubscribeTopic": "",
```

```

"MQTT_KeepaliveTime": 30,
"MQTT_UseGZIP": 0,
"UseWebSocketClient": 0,
"WebSocketClient_ProtocolType": 100,
"WebSocketClient_ServerAddr": "ws://192.168.1.100/ws",
"WebSocketClient_UseGZIP": 0,
"WebSocketClient_KeepaliveTime": 30,
"UseYZW": 0,
"YZWAddr": "http://192.168.1.10",
"RecordAutoCycle": 0,
"SaveUnregistered": 1,
"SaveRecordPicture": 1,
"DeviceName": "",
"Manufacturer": "",
"ManufacturerPhone": "",
"Website": "",
"ProductionDate": "2024-06-15",
"OEMText": "",
"AutoRestart": 0,
"AutoRestartTime": "02:00",
"TimeGroups": [
  {
    "Num": 1,
    "Week1": "00:00-23:59",
    "Week2": "00:00-23:59",
    "Week3": "00:00-23:59",
    "Week4": "00:00-23:59",
    "Week5": "00:00-23:59",
    "Week6": "00:00-23:59",
    "Week7": "00:00-23:59"
  },
  {
    "Num": 64,
    "Week1": "",
    "Week7": ""
  }
],
"DisplayBrightness": 6,
"MenuPassword": "0000",
"ShowIR": 0,
"ShowPersonPhoto": 1,
"PlayPersonName": 1,
"RecognitionButon": 0,
"UnregisteredWarn": 0,
"ShowPersonName": 1,
"FillLight": 2
}

```

## Working parameters received on device

### Brief description:

- After receiving the \* \* PushWorkSetting \* \* message sent by the server, the device saves the parameters and replies to the server for confirmation that the device has received them



## Publisher

- Device

## Command

- PushWorkSettingACK

## Message parameters

| Field | Type   | Explain                                                |
|-------|--------|--------------------------------------------------------|
| Cmd   | string | Command type                                           |
| CmdID | string | Command ID, the ID in the command issued by the server |
| Body  | object | Results saved in device                                |

## Results saved in device

| Parameter Name | Type   | Necessary | Description                         |
|----------------|--------|-----------|-------------------------------------|
| Result         | int    | Yes       | 1-- Save successful, 0--Save failed |
| ErrorMsg       | string | No        | Error Message                       |

## Message Example

```
{
  "cmd": "PushWorkSettingACK",
  "CmdTime":123456789, //unix timestamp, representing the time of the command
  "CmdID":"123456",
  "Body":{
    "Result" : 1,
    "ErrorMsg" : ""
  }
}
```

# Remote control

## The server sends remote operation instructions

### Brief description:

- When the server needs devices to execute remote commands, the server actively issues them

## Publisher

- Server

## Command

- RemoteCommand

## Message parameters

| Field | Type   | Explain                                                |
|-------|--------|--------------------------------------------------------|
| Cmd   | string | Command type                                           |
| CmdID | string | Command ID, the ID in the command issued by the server |
| Body  | object | Instruction structure that requires instructions       |

## Remote operation instruction structure

| Parameter Name | Type   | Necessary | Description                                                                                                                                                                                                                   |
|----------------|--------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Restart        | int    | No        | Remote restart ``0:no restart, 1:restart                                                                                                                                                                                      |
| Recover        | int    | No        | Factory reset ``0:normal, 1: Factory reset                                                                                                                                                                                    |
| Opendoor       | int    | No        | Remote door opening command ` 0: Not processed,<br>1--Turn on relay; 2--Keep the door normally open; 3--Close the door (removed normally open);<br>` 4--Lock the door;5--Unlock the door                                      |
| Closealarm     | int    | No        | Close alarm command ``0: Do not process, 1:Close all ongoing alarms and record them                                                                                                                                           |
| RepostRecord   | int    | No        | Re upload records ``0:Do not process, 1: Mark all uploaded records as not uploaded and resend                                                                                                                                 |
| PushAllPeople  | int    | No        | Request to upload all stored personnel lists to the server<br>At this time, the device calls the publish message[UploadPeople] to send the personnel list<br>0--No processing required, 1--All personnel need to be uploaded. |
| QueryPeople    | [uint] | No        | Request to upload personnel with the specified user ID to the server<br>At this time, the device calls to publish a message[UploadPeople]and sends a list of personnel.<br>The type is an array                               |
| ClearRecord    | int    | No        | Delete all records; 0-- Do not process 1--Delete all records                                                                                                                                                                  |

## Message Example

```
{
  "cmd": "RemoteCommand",
  "CmdTime": 123456789, //unix timestamp, representing the time of the command
  was issued
  "CmdID": "1234567",
  "body":
  {
    "opendoor": 1,
    "QueryPeople": [1,2,3,4,5,6]
  }
}
```

## Device confirms receipt of remote command

### Brief description:

- When the device receives the remote command \* \* RemoteCommand \* \* sent by the server and completes the execution according to the command content, the device will issue this confirmation message to the server

### Publisher

- Device

### Command

- RemoteCommandACK

### Message Parameters

| Field | Type   | Explain                                                |
|-------|--------|--------------------------------------------------------|
| Cmd   | string | Command type                                           |
| CmdID | string | Command ID, the ID in the command issued by the server |

## Message Example

```
{
  "cmd": "RemoteCommandACK",
  "CmdTime": 123456789, //Unix timestamp, representing the time of the command
  was issued
  "CmdID": "123456"
}
```

#

# Personnel Management

## Server push personnel

### Brief description:

- When there are changes in the permissions or information of server-side personnel that need to be synchronized to the device, the server sends this message
- A message only supports one personnel information, do not send personnel authorization information continuously. The device is single threaded processing, and continuous sending will result in packet loss

### Publisher

- Server

### Command

- PushPeople

### Message parameters

| Field | Type     | Explain                                                                                                                                                                                                                                                                                                                     |
|-------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cmd   | string   | Command type                                                                                                                                                                                                                                                                                                                |
| CmdID | string   | Command ID, the ID in the command issued by the server                                                                                                                                                                                                                                                                      |
| body  | [object] | Personnel information list <b>array structure</b><br>If personnel photos are directly carried in the command, only one person can be transmitted at a time<br>If personnel photo data is not carried in the command, only personnel photo URLs can be carried, and a maximum of 100 personnel can be transmitted at a time. |

### PeopleJson Personnel data format

| Parameter Name | Type   | Necessary | Description                                                      |
|----------------|--------|-----------|------------------------------------------------------------------|
| UserID         | int    | Yes       | User ID (number maximum value 4294967295 type UINT32)            |
| Name           | string | No        | Personnel name (characters<64)                                   |
| Job            | string | No        | Position (characters<64)                                         |
| Department     | string | No        | Department (characters<64 bits)                                  |
| IdentityCard   | string | No        | Identification card number (can be blank) (characters<64 digits) |

| Parameter Name | Type   | Necessary | Description                                                                                                                                                                                                                                                                                                                                          |
|----------------|--------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Attachment     | string | No        | Other personnel information (characters<200)                                                                                                                                                                                                                                                                                                         |
| Photo          | string |           | Photo file name, can be empty,<br>The device supports using base64 encoding for photos, but it is not recommended which will increase the traffic. Photos should be transferred using binary files<br>If the file information starts with http or https, it means the photo is a URL address, and the device will download the photo through the URL |
| PhotoMD5       | string | No        | MD5 HEX string format for photos (characters=32 digits)                                                                                                                                                                                                                                                                                              |
| PhotoLen       | int    | No        | The maximum length of the photo supported is 400KB                                                                                                                                                                                                                                                                                                   |
| Password       | string | No        | Password, only number, length: (0 / 4-8)                                                                                                                                                                                                                                                                                                             |
| CardNum        | string | No        | Card number (digits, maximum value 1844674407909551615 type UINT62)                                                                                                                                                                                                                                                                                  |
| QRCode         | string | No        | Personnel QR code information (characters<128)                                                                                                                                                                                                                                                                                                       |
| AccessType     | int    | No        | Role 0--Ordinary personnel; 1--Administrator; 2--Blacklist                                                                                                                                                                                                                                                                                           |
| ExpirationDate | uint32 | No        | Permission expiration date<br>unix Unix timestamp in seconds 0 indicates No time limit<br>Maximum indicates December 31, 2099                                                                                                                                                                                                                        |
| OpenTimes      | int    | No        | Opening Times 0-65535;<br>65535--Indicates no restrictions, 0--indicates no passage allowed                                                                                                                                                                                                                                                          |
| KeepOpen       | int    | No        | Whether is the normally opened card, 1--Yes; 0--No                                                                                                                                                                                                                                                                                                   |
| Timegroup      | int    | No        | Opening time zone group 1-64; 0--indicates restricted                                                                                                                                                                                                                                                                                                |
| Holidays       | string | No        | Holiday restrictions<br>comma separated: 1,2,3,4,5                                                                                                                                                                                                                                                                                                   |
| Elevators      | string | No        | Elevator port permission array<br>comma separate: 1,2,3,4,5                                                                                                                                                                                                                                                                                          |

| Parameter Name | Type   | Necessary | Description                                                                                                                                                                                                          |
|----------------|--------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FaceFeature    | string | No        | Facial feature codes starting with http or https indicate the use of a URL address, otherwise they indicate the use of base64. Download the feature code through the file, and the downloaded file content is base64 |
| FaceFeatureMD5 | string | No        | MD5 value HEX string format of facial feature code                                                                                                                                                                   |
| Fingerprints   | []     | No        | Fingerprint Object                                                                                                                                                                                                   |
| Palmveins      | []     | No        | Palm print object                                                                                                                                                                                                    |

**Characters refer to bytes, with one Chinese character occupying 3-4 bytes (UTF-8 encoding) and one English character occupying 1 byte.**

#### Holidays Holiday

- An empty string or no field indicates no holiday restrictions
- For specific restricted holiday numbers, comma separated: 1,2,3,4,5
- \*Number indicates that it is subject to all holiday restrictions

#### • Fingerprint Fingerprint Object

| Serial number | Type   | Required | description                                                                                                                                                                                                                    |
|---------------|--------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Num           | int    | Yes      | Fingerprint index number                                                                                                                                                                                                       |
| Data          | string | Yes      | The fingerprint feature code starting with http or https indicates the use of a URL address, otherwise it indicates the use of base64<br>Download the feature code through the file, and the downloaded file content is base64 |
| MD5           | string | No       | MD5 value of feature code in HEX string format                                                                                                                                                                                 |

```
[ //Structural Example
  {
    Num: 1,
    Data: "base64 code",
    MD5: "abcdefg"
  },
  {
    Num: 2,
    Data: "base64 code",
    MD5: "abcdefg"
  }
]
```

- **Palmvein Palm print object**

| Serial number | Type   | Required | Description                                                                                                                                                                                                                     |
|---------------|--------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Num           | int    | Yes      | Palm print index number                                                                                                                                                                                                         |
| Data          | string | Yes      | The palm print feature code starting with http or https indicates the use of a URL address, otherwise it indicates the use of base64<br>Download the feature code from a file, and the content of the downloaded file is base64 |
| MD5           | string | Np       | MD5 value of the feature code HEX string format                                                                                                                                                                                 |

```
[ //Structural Example
  {
    Num: 1,
    Data: "http://www.abc.com/1.txt",
    MD5: "abcdefg"
  },
  {
    Num: 2,
    Data: "base64 code",
    MD5: "abcdefg"
  }
]
```

- **Elevator Elevator permissions**

```
//This person has elevator permissions for floors 1-5
[
  1,2,3,4,5
]
//This person does not have elevator permission
[
]
//This person only has access permission on the 10th floor
[
  10
]
```

## Message Example

```
{
  "cmd": "PushPeople",
  "CmdID": "6666",
  "body": [
    {
      "UserID": "3",
```

```

        "Name": "888888",
        "Job": "development",
        "Department": "Sales Department",
        "IdentityCard": "",
        "Attachment": "",
        "Photo": "1.jpg",
        "PhotoMD5": "613D870CA99EDF074BEE4387BAB09070",
        "PhotoLen": 55020,
        "Password": "2222",
        "CardNum": "6666",
        "AccessType": 0,
        "ExpirationDate": 0,
        "OpenTimes": 65535,
        "KeepOpen": 1,
        "Timegroup": 6,
        "Holidays": "1,3,9,10,11,17,21,25,27,30",
        "Elevators": "1,2,3,4,5",
        "FaceFeature": "base64 code",
        "FaceFeatureMD5": "613D870CA99EDF074BEE4387BAB09070",
        "Fingerprints": [
            {
                Num: 1,
                Data: "base64 code",
                MD5: "abcdefg"
            }
        ],
        "Palmveins": [
            {
                Num: 1,
                Data: "base64 code",
                MD5: "abcdefg"
            }
        ]
    }
}

```

## **\*\* Device feedback personnel store results\*\***

### **Brief description:**

- After the device receives the personnel sent by the server and saves it, then send this message to notify the server personnel to store the results

### **Publisher**

- Device



## Command

- PushPeopleACK

## Message parameters

| Field | Type   | Explain                                                |
|-------|--------|--------------------------------------------------------|
| Cmd   | string | Command type                                           |
| CmdID | string | Command ID, the ID in the command issued by the server |
| body  | object | Personnel storage result feedback structure            |

### Personnel storage result feedback structure

| Parameter Name | Type     | Necessary | Description                     |
|----------------|----------|-----------|---------------------------------|
| SuccessCount   | int      | Yes       | Number of successful imports    |
| FailCount      | int      | Yes       | Number of failed imports        |
| FailList       | [object] | No        | Import failed personnel details |

### Import failed personnel structure

| Parameter Name | Type   | Necessary | Description                                                               |
|----------------|--------|-----------|---------------------------------------------------------------------------|
| UserID         | string | Yes       | Import user ID failed (number maximum value 4294967295 type UINT32)       |
| ErrorCode      | int    | Yes       | Error code                                                                |
| RepeatID       | string | No        | Duplicate user ID, this field indicates who this person is repeating with |
| ErrMsg         | string | No        | Error description                                                         |

## Message Example

```
{
  "cmd": "PushPeople",
  "CmdTime":123456789,//unix timestamp, representing the time of the command
was issued
  "CmdID":"6666",
  "body":{
    "SuccessCount" : 1,
    "FailCount" : 1,
    "FailList" : {
      "UserID":1234,
      "ErrorCode":20,
      "RepeatID":2222,
      "ErrMsg" : "Duplicate facial feature codes"
    }
  }
}
```

```
}  
}
```

### Personnel failure error code

| Success | description                                                                                               |
|---------|-----------------------------------------------------------------------------------------------------------|
| 1       | Abnormal personnel information parameters, specific reasons refer to                                      |
| 2       | Personnel photo URL download failed                                                                       |
| 3       | The personnel photo image is too large and does not meet the requirements. It needs to be less than 500kb |
| 4       | Facial feature code download failed                                                                       |
| 5       | Fingerprint feature code download failed                                                                  |
| 6       | Palm vein feature code download failed                                                                    |
| 20      | Duplicate facial feature codes                                                                            |
| 21      | Duplicate fingerprint feature codes                                                                       |
| 22      | Duplicate palm vein feature code                                                                          |
| 23      | Duplicate card number                                                                                     |
| 24      | Personnel are full                                                                                        |
| 25      | An error occurred while querying data                                                                     |
| 26      | Error occurred while saving personnel photos                                                              |
| 27      | Facial feature code format error                                                                          |
| 28      | Fingerprint feature code format error                                                                     |
| 29      | Palm vein feature code format error                                                                       |
| 30      | Photo not recognized                                                                                      |
| 31      | Error occurred while saving fingerprint feature code                                                      |
| 32      | Error occurred while saving palm vein feature code                                                        |

## The server sends a message to delete personnel

### Brief description:

- When the server personnel permissions change and need to be deleted, the server actively sends this message
- Since deleting personnel only requires sending the personnel user number, it can support sending multiple personnel at a time

## Publisher

- Server

## Command

- PushDeletePeople

## Message parameters

| Field | Type   | Expalin                                                |
|-------|--------|--------------------------------------------------------|
| Cmd   | string | Command type                                           |
| CmdID | string | Command ID, the ID in the command issued by the server |
| body  | object | Personnel parameter structure to be deleted            |

- Personnel parameter structure to be deleted

| Parameter Name | Type | Necessary | Description                                                                     |
|----------------|------|-----------|---------------------------------------------------------------------------------|
| DeleteAll      | int  | No        | 1: Clear all personnel information 0: Delete according to the specified user ID |
| DeleteCount    | int  | No        | Number of personnel to be deleted                                               |
| DeleteList     | []   | No        | When DeleteAll=0, this parameter contains the user ID that needs to be deleted  |

## Message Example

```
{
  "cmd": "PushDeletePeople",
  "CmdTime":123456789,//unix timestamp, representing the time of the command
was issued
  "CmdID":"6666",
  "body":{
    "DeleteAll":0,
    "DeleteCount": 5,
    "DeleteList":[
      1,2,3,4,5
    ]
  }
}
```

## Device feedback on personnel deletion results

Brief description:

- When the device receives a message from the server to delete personnel and completes the deletion, it sends this message to notify the server of the result of deleting personnel

## Publisher

- Device

## Command

- PushDeletePeopleACK

## Message parameters

| Field | Type   | Explain                                                |
|-------|--------|--------------------------------------------------------|
| Cmd   | string | Command type                                           |
| CmdID | string | Command ID, the ID in the command issued by the server |
| body  | object | Deleted personnel result                               |

- Personnel parameter structure to be deleted

| Parameter Name | Type | Necessary | Description                                                   |
|----------------|------|-----------|---------------------------------------------------------------|
| DeleteAll      | int  | No        | Have all personnel been cleared? 1: Yes 0: No                 |
| DeleteCount    | int  | No        | Number of personnel deleted                                   |
| DeleteList     | []   | No        | When DeleteAll=0, this parameter contains the deleted user ID |

## Message Example

```
{
  "cmd": "PushDeletePeopleACK",
  "CmdTime":123456789,//Unix timestamp, representing the time of the command
was issued
  "CmdID":"6666",
  "body":{
    "DeleteAll":0,
    "DeleteCount": 5,
    "DeleteList":[
      1,2,3,4,5
    ]
  }
}
```

# Device pushes personnel information

## Brief description:

- When personnel are added or modified on the device, the personnel whose device changes will be sent to the server
- When the server requests to upload personnel information, send the personnel information to the server
- Push personnel information to the server when requested to upload all personnel information

## Publisher

- Device

## Command

- UploadPeople

## Message parameters

| Field | Type   | Explain                                                                                         |
|-------|--------|-------------------------------------------------------------------------------------------------|
| Cmd   | string | Command type                                                                                    |
| CmdID | string | Command ID, the command ID generated by the device needs to be carried when the server responds |
| body  | object | Modified personnel information structure                                                        |

## Personnel Information Structure

| Field    | Type  | Length | Required | Explain                                                                                                 |
|----------|-------|--------|----------|---------------------------------------------------------------------------------------------------------|
| PushType | int   | 1      | Yes      | Types of personnel changes in device: 1--New; 2--Update; 3--Delete; 4--Query;                           |
| UserID   | uint  |        | Yes      | Personnel User ID                                                                                       |
| Detail   | class |        | No       | Personnel details, if the personnel do not exist or the type is deleted, then this field does not exist |

## Message Example

```
{
  "cmd": "UploadPeople",
  "CmdTime":123456789,//unix timestamp, representing the time of the command
  was issued
}
```

```

"CmdID": "6666",
"body": {
  "PushType": 1,
  "UserID": "3",
  "Detail": {
    "UserID": "3",
    "Name": "888888",
    "Job": "development",
    "Department": "Sales Department",
    "IdentityCard": "",
    "Attachment": "",
    "Photo": "1.jpg",
    "PhotoMD5": "613D870CA99EDF074BEE4387BAB09070",
    "PhotoLen": 55020,
    "Password": "2222",
    "CardNum": "6666",
    "AccessType": 0,
    "ExpirationDate": 0,
    "OpenTimes": 65535,
    "KeepOpen": 1,
    "Timegroup": 6,
    "Holidays": "1,3,9,10,11,17,21,25,27,30",
    "Elevators": "1,2,3,4,5",
    "FaceFeature": "base64 code",
    "FaceFeatureMD5": "613D870CA99EDF074BEE4387BAB09070",
    "Fingerprints": [
      {
        Num: 1,
        Data: "base64 code",
        MD5: "abcdefg"
      }
    ],
    "Palmveins": [
      {
        Num: 1,
        Data: "base64 code",
        MD5: "abcdefg"
      }
    ]
  }
}
}

```

## ***\*Server feedback received personnel pushed by devices***

---

### **Brief description:**

- When the server receives the **UploadPeople** When the server receives the, it should issue the **UploadPeopleACK** message to confirm that the device has received it and continue to send the next person

## Publisher

- Server

## Command

- UploadPeopleACK

## Message parameters

| Field | Type   | Explain                                                |
|-------|--------|--------------------------------------------------------|
| Cmd   | string | Command type                                           |
| CmdID | string | Command ID, the ID in the command issued by the server |

## Message Example

```
{
  "cmd": "UploadPeopleACK",
  "CmdTime":123456789,//unix timestamp, representing the time of the command
was issued
  "CmdID":"6666"
}
```

##

# Record management

## Upload punch-in records on the device

### Brief description:

- When there is a new punch-in record in the device, the device will release a new record message
- Due to the punch-in records will include on-site photos, this message only uploads one punch-in record at a time

## Publisher

- Device

## Command

- UploadIdentifyRecord

## Message parameters

| Field | Type   | Explain                                                                                         |
|-------|--------|-------------------------------------------------------------------------------------------------|
| Cmd   | string | Command type                                                                                    |
| CmdID | string | Command ID, the command ID generated by the device needs to be carried when the server responds |
| body  | object | Punch-in record structure                                                                       |

## Punch-in record structure

| Field        | Type   | Length | Required | Explain                                                      |
|--------------|--------|--------|----------|--------------------------------------------------------------|
| RecordID     | long   | 10     | Yes      | Record number                                                |
| RecordType   | int    | 5      | Yes      | Event type                                                   |
| RecordDate   | int64  | 20     | Yes      | Punch-in time unix timestamp                                 |
| UserID       | string | 20     | No       | User ID                                                      |
| Name         | string | 30     | No       | Personnel Name                                               |
| IdentityCard | string | 5      | No       | Identification Card                                          |
| Job          | string | 1      | No       | Position                                                     |
| Department   | string | 1      | No       | Department                                                   |
| CardNum      | string | 20     | No       | Card No.                                                     |
| QRCode       | string | 128    | No       | QR code                                                      |
| IsEntry      | int    | 1      | No       | Is it entry入, 1 indicates entry, 0 indicates exit            |
| BodyTemp     | int    | 5      | No       | Human body temperature measurement requires a division of 10 |
| PhotoLen     | int    | 10     | No       | Image file length 0 indicates no image                       |

## RecordType Event Type

| Value | Explain                  |
|-------|--------------------------|
| 1     | Card verification        |
| 2     | Fingerprint verification |
| 3     | Facial verification      |
| 4     | Card + Fingerprint       |



| Value | Explain                                                                                                                                                  |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5     | Face + Fingerprint                                                                                                                                       |
| 6     | Card + Face                                                                                                                                              |
| 7     | Card + PIN                                                                                                                                               |
| 8     | Face + PIN                                                                                                                                               |
| 9     | Fingerprint + PIN                                                                                                                                        |
| 10    | Password verification user number plus password                                                                                                          |
| 11    | Card + Fingerprint + PIN                                                                                                                                 |
| 12    | Card + Face + PIN                                                                                                                                        |
| 13    | Fingerprint + Face + PIN                                                                                                                                 |
| 14    | Card + Fingerprint + Face                                                                                                                                |
| 15    | Repeated verification                                                                                                                                    |
| 16    | Expired validity period                                                                                                                                  |
| 17    | Opening time zone expired                                                                                                                                |
| 18    | Cannot open the door during holidays                                                                                                                     |
| 19    | Unregistered user                                                                                                                                        |
| 20    | Detection lock                                                                                                                                           |
| 21    | The number of valid times has been exhausted                                                                                                             |
| 22    | Verify when locked, prohibit opening the door                                                                                                            |
| 23    | Lost reported card                                                                                                                                       |
| 24    | Blacklist Card                                                                                                                                           |
| 25    | Open the door without verification -- when pressing the fingerprint, the user number is 0, and when swiping the card, the user number is the card number |
| 26    | Prohibit card swiping verification -- When card swiping is disabled in the [Permission Authentication Mode]                                              |
| 27    | Prohibit fingerprint verification -- [Permission authentication mode] When fingerprint is disabled                                                       |
| 28    | Controller has expired                                                                                                                                   |
| 29    | Verified—Validity period is ready to expire                                                                                                              |
| 30    | Abnormal body temperature, refusal to enter                                                                                                              |
| 31    | Visitor password to open the door                                                                                                                        |

| Value | Explain                                               |
|-------|-------------------------------------------------------|
| 32    | Dynamic QR code scanning to open the door             |
| 33    | Add personnel on the device menu                      |
| 34    | Modify personnel on the device menu                   |
| 35    | Delete personnel from the device menu                 |
| 36    | Palm vein recognition                                 |
| 37    | Card + Palm Vein + Face                               |
| 38    | Palm vein recognition + PIN                           |
| 39    | Card + Palm Vein                                      |
| 40    | Face + Palm Vein                                      |
| 41    | Card + Palm Vein + PIN                                |
| 42    | Palm Vein + Face + PIN                                |
| 43    | Fingerprint + Palm Vein + Face                        |
| 44    | Combination verification--waiting for the next person |
| 45    | Combination verification failed                       |
| 46    | Combination verification success                      |
| 47    | Personnel and identification card comparison          |
| 48    | Unregistered card                                     |
| 49    | Unregistered QR code                                  |

## Message Example

```
{
  "cmd": "UploadPeopleACK",
  "CmdTime":123456789,//unix timestamp, representing the time of the command
  "CmdID":"6666",
  "body":{
    "RecordID": 120,
    "RecordType": 3,
    "RecordDate": 1718616771,
    "UserID": "1",
    "Name": "1",
    "IdentityCard": "",
    "Job": "",
    "Department": "",
    "CardNum": "0",
    "QRCode": ""
  }
}
```

```
"IsEntry": 1,
"BodyTemp": 0,
"PhotoLen": 40363 //Indicates that have photos, =0 Indicates that no
photos
}
}
```

## Server feedback received punch-in records pushed by devices

---

### Brief description:

- When the server receives the **UploadIdentifyRecord** command sent by the device, it should issue the **UploadIdentifyRecordACK** message to confirm that the device has received it and continue to send the next record

### Publisher

- Server

### Command

- UploadIdentifyRecordACK

### Message parameters

| Field | Type   | Explain                                                |
|-------|--------|--------------------------------------------------------|
| Cmd   | string | Command type                                           |
| CmdID | string | Command ID, the ID in the command issued by the server |

### Message Example

```
{
  "cmd": "UploadIdentifyRecordACK",
  "CmdTime":123456789,//unix timestamp, representing the time when the command
was issued
  "CmdID":"6666"
}
```

## Device upload system records

---

### Brief description:

- Send this message when there is a new system record in the device

- Due to the lack of on-site photos in the system records, this message will be uploaded in bulk to the system records

## Publisher

- Device

## Command

- UploadSystemRecord

## Message parameters

| Field | Type   | Explain                                                                                         |
|-------|--------|-------------------------------------------------------------------------------------------------|
| Cmd   | string | Command type                                                                                    |
| CmdID | string | Command ID, the command ID generated by the device needs to be carried when the server responds |
| body  | object | System record structure                                                                         |

## System record structure

| Field      | Type     | Length | Required | Explain                                               |
|------------|----------|--------|----------|-------------------------------------------------------|
| RecordType | int      | 1      | Yes      | Event type<br>1-Door magnetic record;2--System Record |
| Records    | [object] |        | Yes      | System record list                                    |

## Records Field Description

| Field      | Type  | Length | Required | Explain                                               |
|------------|-------|--------|----------|-------------------------------------------------------|
| RecordID   | long  | 10     | Yes      | Record number                                         |
| RecordType | int   | 5      | Yes      | Event type<br>1-Door magnetic record;2--System Record |
| RecordDate | int64 | 20     | Yes      | Punch-in time unix timestamp                          |

## RecordType Field Description

| Value | Explain              |
|-------|----------------------|
| 1     | Door magnetic record |
| 2     | System Record        |

## System record structure

| Field      | Type     | Length | Required | Explain                                               |
|------------|----------|--------|----------|-------------------------------------------------------|
| RecordType | int      | 1      | Yes      | Event type<br>1-Door magnetic record;2--System Record |
| Records    | [object] |        | Yes      | System Record List                                    |

#####

### RecordType -- Door magnetic record Event type

| Value | Explain                                                                      |
|-------|------------------------------------------------------------------------------|
| 1     | Door Magnetic-Open Door                                                      |
| 2     | Door Magnetic-Close Door                                                     |
| 3     | Enter the door magnetic alarm detection state                                |
| 4     | Exit the door magnetic alarm detection state                                 |
| 5     | The door is not completely closed                                            |
| 6     | Exit button to open the door                                                 |
| 7     | When the exit button is pressed to open the door, the door is already locked |
| 8     | When the exit button is pressed to open the door, the controller has expired |

### RecordType -- System records event type

| Value | Explain                                                |
|-------|--------------------------------------------------------|
| 1     | Software Open Door                                     |
| 2     | Software Close Door                                    |
| 3     | Software Normally opening                              |
| 4     | The controller automatically enters normally open mode |
| 5     | The controller automatically closes the door           |

| Value | Explain                                                        |
|-------|----------------------------------------------------------------|
| 6     | Press and hold the exit button to make normally open           |
| 7     | Press and hold the exit button to make normally closed         |
| 8     | Software locked                                                |
| 9     | Software removed locked                                        |
| 10    | Controller timing locked--automatic locking at specific time   |
| 11    | Controller timing locked--automatic unlocking at specific time |
| 12    | Alarm--Locked                                                  |
| 13    | Alarm--Unlock                                                  |
| 14    | Illegal authentication alarm                                   |
| 15    | Door magnetic alarm                                            |
| 16    | Duress Alarm                                                   |
| 17    | Timeout alarm of open door                                     |
| 18    | Blacklist alarm                                                |
| 19    | Fire alarm                                                     |
| 20    | Anti disassembly alarm                                         |
| 21    | Illegal authentication alarm removed                           |
| 22    | Door magnetic alarm removed                                    |
| 23    | Duress alarm removed                                           |
| 24    | Timeout alarm of open door removed                             |
| 25    | Blacklist alarm removed                                        |
| 26    | Fire Alarm removed                                             |
| 27    | Anti disassembly alarm removed                                 |
| 28    | System power-on                                                |
| 29    | System error reset (watchdog)                                  |
| 30    | Device formatting record                                       |
| 31    | The wiring of the card reader is reversed                      |
| 32    | The card reader circuit is not properly connected              |
| 33    | Unrecognized card reader                                       |
| 34    | The network cable has been disconnected                        |

| Value | Explain                                            |
|-------|----------------------------------------------------|
| 35    | The network cable has been inserted                |
| 36    | WIFI connected                                     |
| 37    | WIFI disconnected                                  |
| 38    | Bluetooth open door                                |
| 39    | Roll call timeout                                  |
| 40    | Clear all personnel from the device menu           |
| 41    | Backup personnel to USB drive in the device menu   |
| 42    | Import personnel from USB drive in the device menu |
| 43    | Remote door opening by indoor unit                 |
| 44    | Delete all records                                 |
| 45    | Delete all personnel                               |

## Message Example

```
{
  "cmd": "UploadSystemRecord",
  "CmdID": "6666",
  "CmdTime": 123456789, //unix timestamp, representing the time of the command
  "body": {
    "RecordType": 1,
    "Records": [
      {
        "RecordID": 1,
        "RecordType": 1,
        "RecordDate": 1718616771
      },
      {
        "RecordID": 2,
        "RecordType": 1,
        "RecordDate": 1718616772
      }
    ]
  }
}
```

# Server feedback received system records pushed by devices

---

## Brief description:

- When the server receives the **UploadSystemRecord** command sent by the device, it should issue a **UploadSystemRecordACK** message to confirm that the device has received it and continue to send the next record

## Publisher

- Server

## Command

- UploadSystemRecordACK

## Message parameters

| Field | Type   | Explain                                                |
|-------|--------|--------------------------------------------------------|
| Cmd   | string | Command type                                           |
| CmdID | string | Command ID, the ID in the command issued by the server |

## Message Example

```
{
  "cmd": "UploadSystemRecordACK",
  "CmdTime":123456789,//Unix timestamp, representing the time of the command
was issued
  "CmdID":"6666"
}
```